Rewriting the definition of conditional probability, we get the 'Multiplication theorem on probability'.

Theorem 12.7

(Multiplication theorem on probability)

The probability of the simultaneous happening of two events A and B is given by

$$P(A \cap B) = P(A/B)P(B)$$
or
$$P(A \cap B) = P(B/A)P(A)$$

12.6.1 Independent Events

Events are said to be independent if occurrence or non-occurrence of any one of the event does not affect the probability of occurrence or non-occurrence of the other events.

Definition 12.15

Two events A and B are said to be independent if and only if

$$P(A \cap B) = P(A) \cdot P(B)$$

Note 12.6

(1) This definition is exactly equivalent to

$$P(A/B) = P(A) \quad \text{if } P(B) > 0$$
$$P(B/A) = P(B) \quad \text{if } P(A) > 0$$

(2) The events $A_1, A_2, A_3, \ldots, A_n$ are mutually independent if
$P(A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2) \cdot \cdots \cdot P(A_n)$.

Theorem 12.8

If A and B are independent then

(i) $\overline{A}$ and $\overline{B}$ are independent.

(ii) A and $\overline{B}$ are independent.

(iii) $\overline{A}$ and B are also independent.

Proof

(i) To prove $\overline{A}$ and $\overline{B}$ are independent:
Since A and B are independent

$$P(A \cap B) = P(A) \cdot (PB)$$

To prove $\overline{A}$ and $\overline{B}$ are independent, we have to prove

$$P(\overline{A} \cap \overline{B}) = P(\overline{A}) \cdot P(\overline{B}).$$

By de Morgan's law

$$P(\overline{A} \cap \overline{B}) = P\left(\overline{A \cup B}\right)$$
$$= 1 - P(A \cup B)$$

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